

Installation et connexion du BlueROV2

BlueROV2 - ROS2 - BlueOS

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Objectifs :

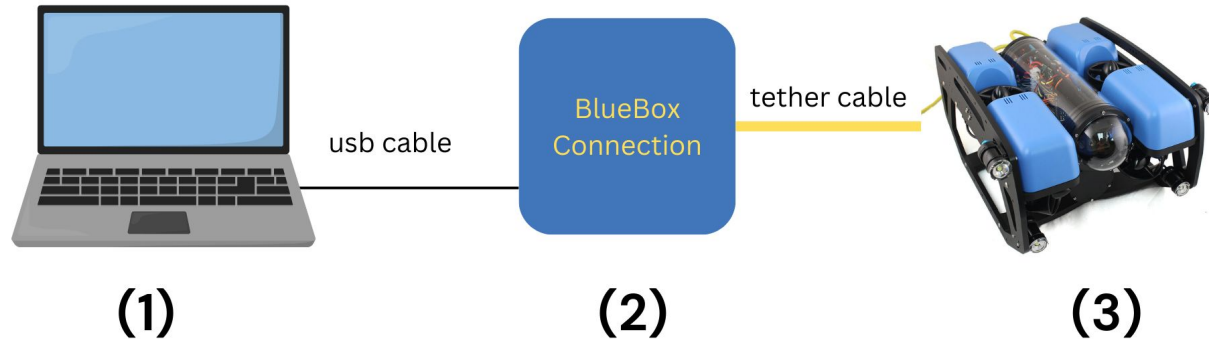
- Architecture du BlueROV2
- Installations nécessaires
- Fonctionnement de ROS2
- Préparation de l'environnement de travail
- Configuration du réseau
- Connexion au robot et lancement de ROS2

Présentation du BlueROV :

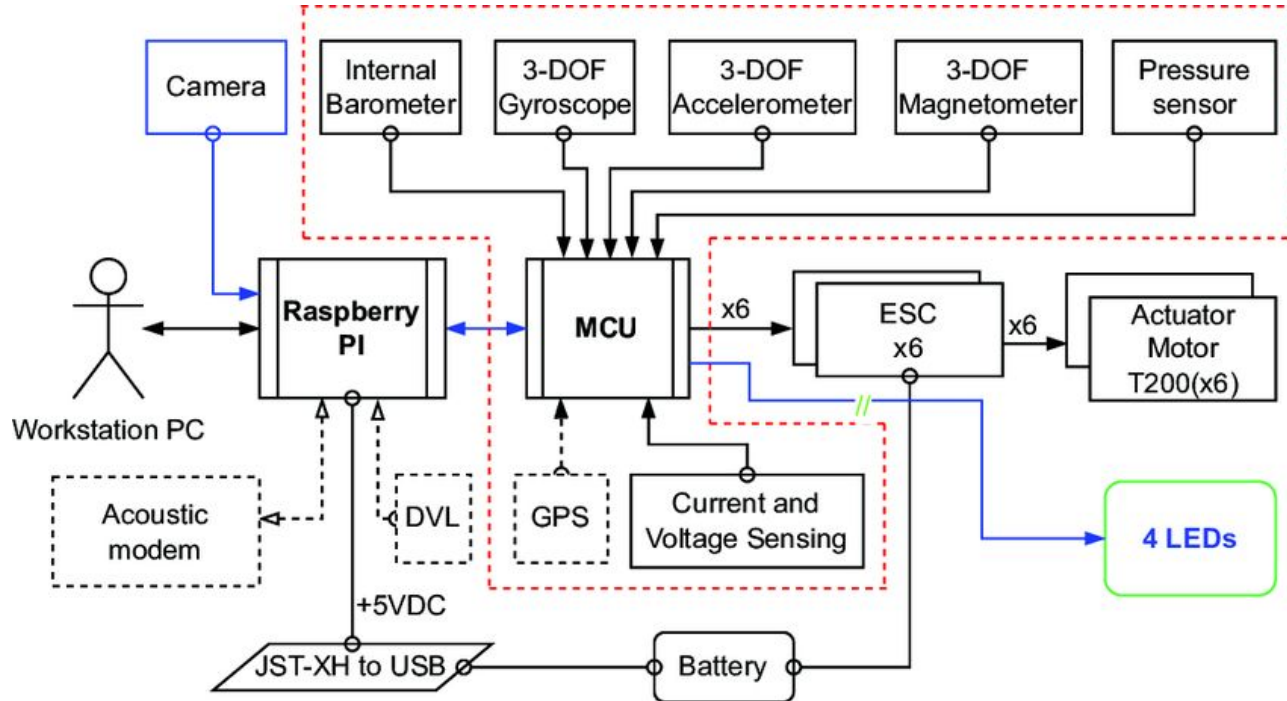


- Robot sous-marin modulaire
- Solution répandue et abordable
- Utilisé en recherche, inspection et expérimentation

Architecture du système :



Architecture interne du système :



Communication avec le robot :



Supervision



Calibration, contrôle,
supervision



Approche
programmable

Prérequis :



Jammy Jellyfish

ROS 2™



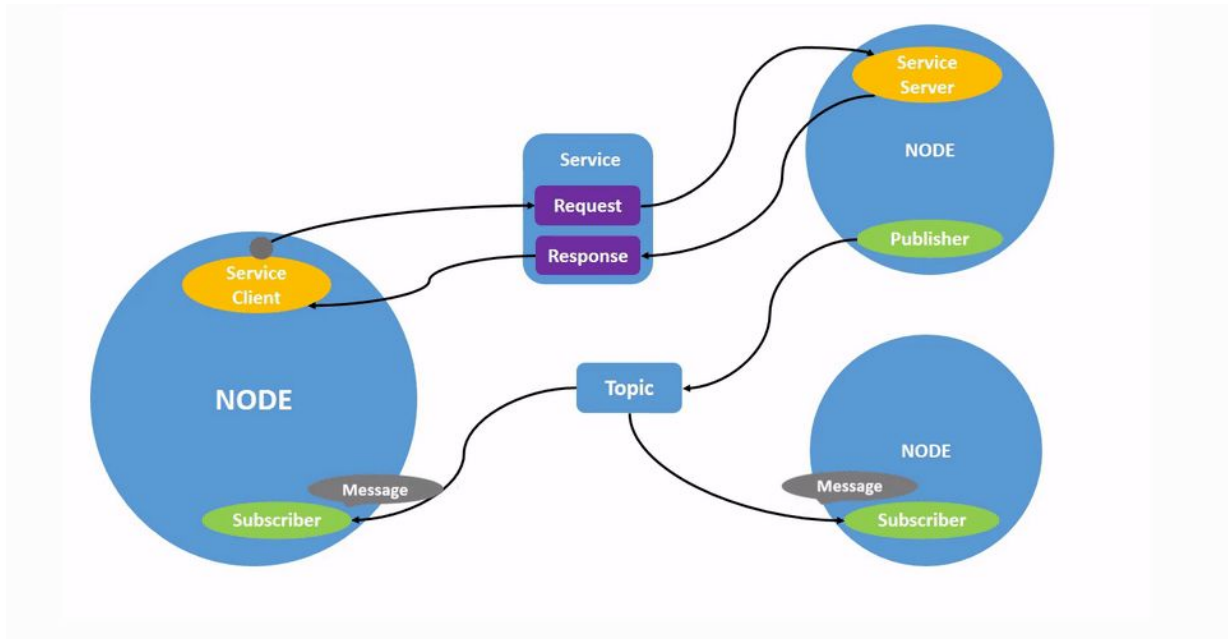
ROS2 Humble

Installation logicielle :



- Ubuntu 22.04
- RO2 Humble
- MAVROS/MAVLINK
- Python + colcon
- QGroundControl
- jstest-gtk
- Terminator

Principe de ROS2

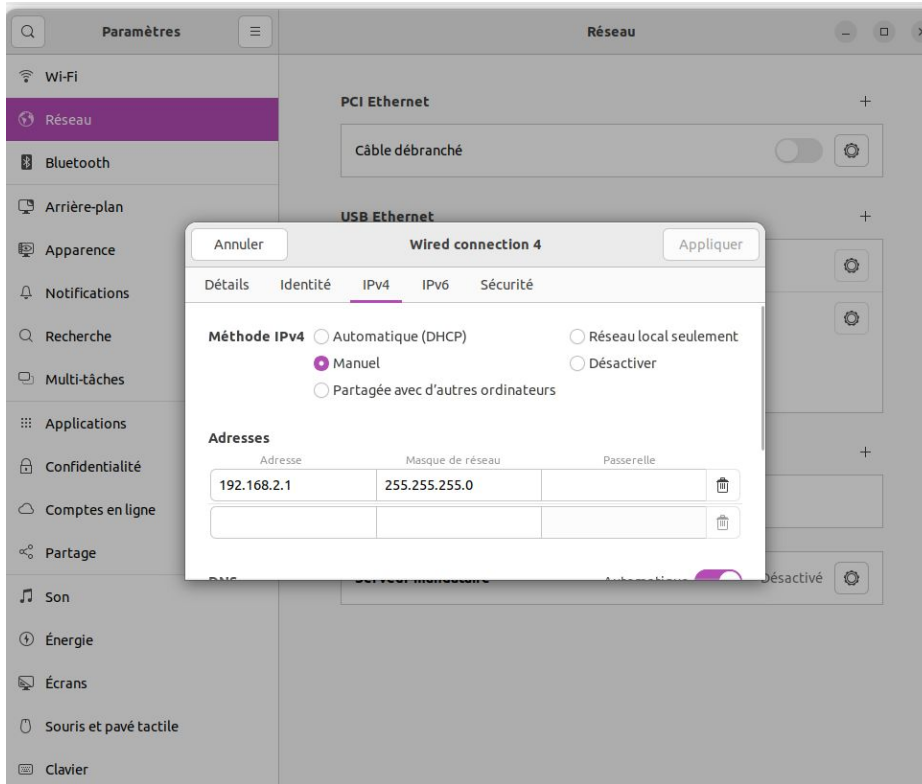


Mise en place workspace :



- Création du workspace
- Clonage du dépôt Github
- Build avec colcon
- Source du workspace

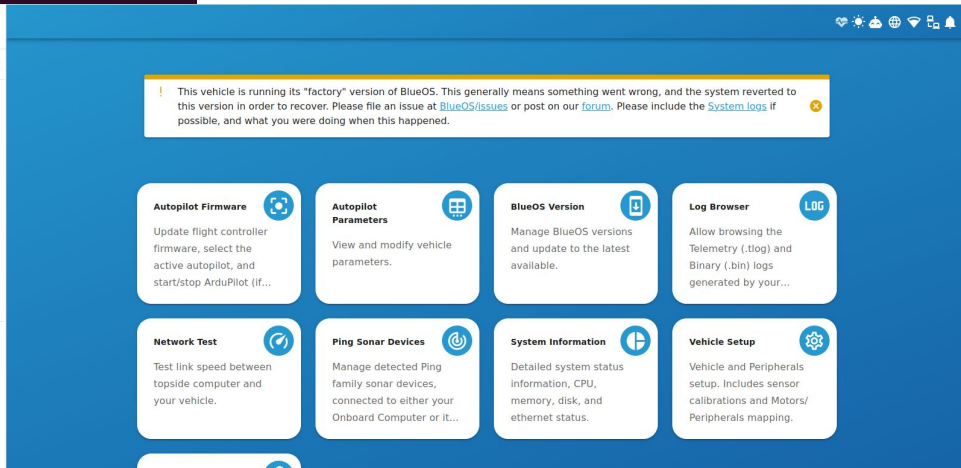
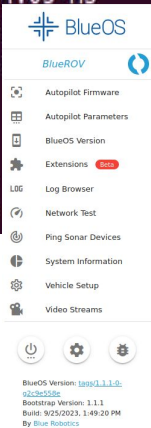
Configuration réseau :



- Création d'un profil dédié
- Configuration manuelle en IPv4
- Adresse : 192.168.2.1
- Masque : 255.255.255.0

Connexion au robot

```
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
64 bytes from 192.168.2.2: icmp_seq=1 ttl=64 time=3.78 ms
64 bytes from 192.168.2.2: icmp_seq=2 ttl=64 time=13.8 ms
64 bytes from 192.168.2.2: icmp_seq=3 ttl=64 time=5.27 ms
64 bytes from 192.168.2.2: icmp_seq=4 ttl=64 time=4.05 ms
64 bytes from 192.168.2.2: icmp_seq=5 ttl=64 time=
64 bytes from 192.168.2.2: icmp_seq=6 ttl=64 time=
^C
--- 192.168.2.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss,
rtt min/avg/max/mdev = 3.608/7.061/13.846/4.157 ms
```



```

ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws 100x27
[mavros_node-1] [WARN] [1774514807.184203830] [bluerov2.time]: TM: RTT too high for timesync: 32.82
ms.
[mavros_node-1] [WARN] [1774514808.288418230] [bluerov2.time]: TM: RTT too high for timesync: 36.68
ms.
[mavros_node-1] [WARN] [1774514809.386560468] [bluerov2.time]: TM: RTT too high for timesync: 33.56
ms.
[mavros_node-1] [WARN] [1774514810.501506561] [bluerov2.time]: TM: RTT too high for timesync: 48.37
ms.
[mavros_node-1] [WARN] [1774514811.582808782] [bluerov2.time]: TM: RTT too high for timesync: 29.55
ms.
[mavros_node-1] [WARN] [1774514812.708891205] [bluerov2.time]: TM: RTT too high for timesync: 56.37
ms.
[mavros_node-1] [WARN] [1774514813.779991734] [bluerov2.time]: TM: RTT too high for timesync: 27.57
ms.
[mavros_node-1] [WARN] [1774514814.897564611] [bluerov2.time]: TM: RTT too high for timesync: 43.92
ms.
[mavros_node-1] [WARN] [1774514815.978221933] [bluerov2.time]: TM: RTT too high for timesync: 24.43
ms.
[mavros_node-1] [WARN] [1774514817.092095478] [bluerov2.time]: TM: RTT too high for timesync: 38.96
ms.
[mavros_node-1] [WARN] [1774514818.376837216] [bluerov2.time]: TM: RTT too high for timesync: 22.68
ms.
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[mavros_node-1] [INFO] [1774514819.208101302] [rclcpp]: signal_handler(SIGINT/SIGTERM)
[INFO] [mavros_node-1]: process has finished cleanly [pid 6068]
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov run_mavros.lau
ch

```

```

ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws 100x27
[teleop_node-2] [INFO] [1774514406.191049143] [TeleopTwistJoy]: Linear axis y on 0 at scale 0.200000
.
[teleop_node-2] [INFO] [1774514406.191052993] [TeleopTwistJoy]: Linear axis z on 4 at scale 0.400000
.
[teleop_node-2] [INFO] [1774514406.191055245] [TeleopTwistJoy]: Angular axis pitch on 6 at scale 0.2
00000.
[teleop_node-2] [INFO] [1774514406.191057536] [TeleopTwistJoy]: Angular axis roll on 7 at scale 0.20
0000.
[teleop_node-2] [INFO] [1774514406.191059626] [TeleopTwistJoy]: Angular axis yaw on 3 at scale 0.200
000.
[joy_node-1] [INFO] [1774514406.580890586] [bluerov2.joy_node]: No haptic (rumble) available, skippi
ng initialization
[joy_node-1] [INFO] [1774514406.581148411] [bluerov2.joy_node]: Opened joystick: Logitech Dual Actio
n. deadzone: 0.200000
[joy_node-1] [INFO] [1774514451.826750328] [bluerov2.joy_node]: Opened joystick: Logitech Gamepad F3
10. deadzone: 0.200000
[joy_node-1] [INFO] [1774514462.880776688] [bluerov2.joy_node]: No haptic (rumble) available, skippi
ng initialization
[joy_node-1] [INFO] [1774514462.880834468] [bluerov2.joy_node]: Opened joystick: Logitech Dual Actio
n. deadzone: 0.200000
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[joy_node-1] [INFO] [1774514821.022768897] [rclcpp]: signal_handler(SIGINT/SIGTERM)
[teleop_node-2] [INFO] [1774514821.022810425] [rclcpp]: signal_handler(SIGINT/SIGTERM)
[INFO] [teleop_node-2]: process has finished cleanly [pid 6230]
[INFO] [joy_node-1]: process has finished cleanly [pid 6228]
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov run_ganepad.lau
ch

```

```

ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws 101x27
[listener-1] KeyboardInterrupt
[listener-1]
[listener-1] During handling of the above exception, another exception occurred:
[listener-1]
[listener-1] Traceback (most recent call last):
[listener-1]   File "/home/ayrton/Documents/bluerov/bluerov_ws/install/autonomous_rov/lib/autonomous_
rov/listener", line 33, in <module>
[listener-1]     sys.exit(load_entry_point('autonomous-rov==0.0.0', 'console_scripts', 'listener'))()
[listener-1]   File "/home/ayrton/Documents/bluerov/bluerov_ws/install/autonomous_rov/lib/python3.10/s
ite-packages/autonomous_rov/listener.py", line 476, in main
[listener-1]     rclpy.shutdown()
[listener-1]   File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/__init__.py", line 130,
in shutdown
[listener-1]     _shutdown(context=context)
[listener-1]   File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/utilities.py", line 58,
in shutdown
[listener-1]     return context.shutdown()
[listener-1]   File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102,
in shutdown
[listener-1]     self._context.shutdown()
[listener-1] rclpy.rclpy_pybind11.RCLError: failed to shutdown: rcl_shutdown already called on the g
iven context, at ./src/rcl/init.c:241
[ERROR] [listener-1]: process has died [pid 6204, exit code 1, cmd '/home/ayrton/Documents/bluerov/bl
uerov_ws/install/autonomous_rov/lib/autonomous_rov/listener --ros-args -r __node:=listener -r __ns:=/
bluerov2'].
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov run_listener.lau
ch

```

```

ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws 101x27
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/utilities.py", line 58, in shutdown
return context.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102, in shutdown
self._context.shutdown()
rclpy.rclpy_pybind11.RCLError: failed to shutdown: rcl_shutdown already called on the given context,
at ./src/rcl/init.c:241
[ros2run]: Process exited with failure 1
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov video --ros-args -r
__node:=video_node
^CTraceback (most recent call last):
File "/home/ayrton/Documents/bluerov/bluerov_ws/install/autonomous_rov/lib/autonomous_rov/video", l
ine 33, in <module>
sys.exit(load_entry_point('autonomous-rov==0.0.0', 'console_scripts', 'video'))()
File "/home/ayrton/Documents/bluerov/bluerov_ws/install/autonomous_rov/lib/python3.10/site-packages
/autonomous_rov/video.py", line 184, in main
rclpy.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/__init__.py", line 130, in shutdown
_shutdown(context=context)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/utilities.py", line 58, in shutdown
return context.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102, in shutdown
self._context.shutdown()
rclpy.rclpy_pybind11.RCLError: failed to shutdown: rcl_shutdown already called on the given context,
at ./src/rcl/init.c:241
[ros2run]: Process exited with failure 1
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 run autonomous_rov video --ros-args -r
__node:=video_node$

```


ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws 10x27

```
[mavros_node-1] [INFO] [1774517938.625880599] [bluerov2.mission]: WP: mission received
[mavros_node-1] [WARN] [1774517939.110914646] [bluerov2.time]: TM: RTT too high for timesync: 15.15
ms.
[mavros_node-1] [WARN] [1774517940.221218712] [bluerov2.time]: TM: RTT too high for timesync: 25.35
ms.
[mavros_node-1] [WARN] [1774517941.321157289] [bluerov2.time]: TM: RTT too high for timesync: 25.38
ms.
[mavros_node-1] [WARN] [1774517942.421232267] [bluerov2.time]: TM: RTT too high for timesync: 25.02
ms.
[mavros_node-1] [WARN] [1774517943.545903949] [bluerov2.time]: TM: RTT too high for timesync: 50.07
ms.
[mavros_node-1] [INFO] [1774517943.621335364] [bluerov2.mission]: WP: mission received
[mavros_node-1] [WARN] [1774517944.630296918] [bluerov2.time]: TM: RTT too high for timesync: 15.15
ms.
[mavros_node-1] [WARN] [1774517945.740723752] [bluerov2.time]: TM: RTT too high for timesync: 25.35
ms.
[mavros_node-1] [WARN] [1774517946.822773500] [bluerov2.time]: TM: RTT too high for timesync: 25.38
ms.
[mavros_node-1] [WARN] [1774517947.911159362] [bluerov2.time]: TM: RTT too high for timesync: 25.02
ms.
[mavros_node-1] [WARN] [1774517949.221235208] [bluerov2.time]: TM: RTT too high for timesync: 50.07
ms.
[mavros_node-1] [WARN] [1774517949.586936794] [bluerov2.time]: TM: RTT too high for timesync: 15.15
ms.
[mavros_node-1] [INFO] [1774517949.621062179] [bluerov2.mission]: WP: mission received
[mavros_node-1] [WARN] [1774517950.324680709] [bluerov2.time]: TM: RTT too high for timesync: 15.15
ms.
```

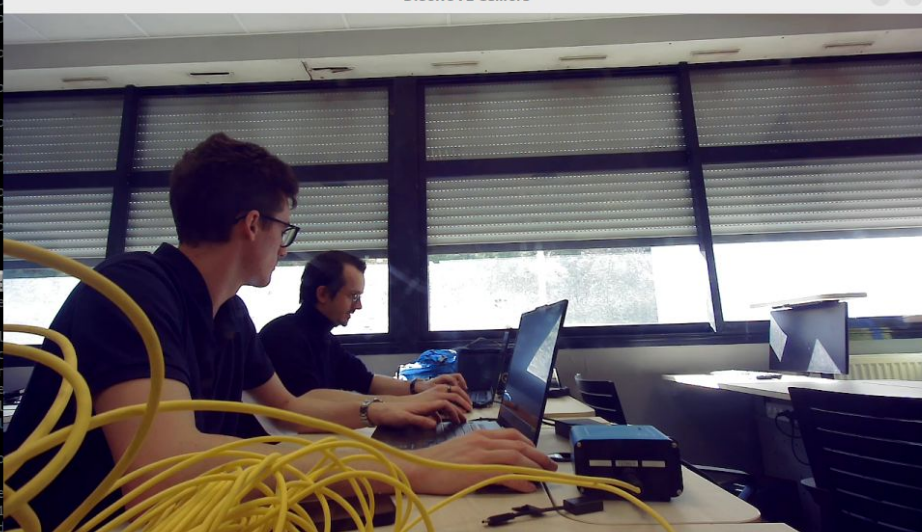
ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws 10x27

```
[joy_node-1] [INFO] [1774514462.880834468] [bluerov2.joy]: deadzone: 0.200000
[C[WARNING] [launch]: user interrupted with ctrl+C
[joy_node-1] [INFO] [1774514821.022768897] [rclcpp_lifecycle]: state is transitioned to TRANSITIONING
[tealeop_node-2] [INFO] [1774514821.022810425] [rclcpp_lifecycle]: state is transitioned to TRANSITIONING
[INFO] [tealeop_node-2]: process has finished cleanly
[INFO] [joy_node-1]: process has finished cleanly
ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov run_listener.launch
[INFO] [launch]: All log files can be found below /home/ayrton/.ros/log/2026-03-26-10-38-41-605604-ayrton-LOQ-15IAX9E-11150
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [joy_node-1]: process started with pid 1774517949
[INFO] [tealeop_node-2]: process started with pid 1774517921
[tealeop_node-2] [INFO] [1774517921.687675965] [TeleopTwistJoy]: Linear axis x on 1 at scale 0.200000
[tealeop_node-2] [INFO] [1774517921.687707653] [TeleopTwistJoy]: Linear axis y on 0 at scale 0.200000
[tealeop_node-2] [INFO] [1774517921.687711381] [TeleopTwistJoy]: Linear axis z on 4 at scale 0.400000
[tealeop_node-2] [INFO] [1774517921.687715959] [TeleopTwistJoy]: Angular axis pitch on 6 at scale 0.200000
[tealeop_node-2] [INFO] [1774517921.687718078] [TeleopTwistJoy]: Angular axis roll on 7 at scale 0.200000
[tealeop_node-2] [INFO] [1774517921.687719927] [TeleopTwistJoy]: Angular axis yaw on 3 at scale 0.200000
```

ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws 10x27

```
[listener-1] File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102,
in shutdown
[listener-1] self.__context.shutdown()
[listener-1] rclpy._rclpy_pybind11.RCLLError: failed to shutdown: rcl_shutdown already called on the g
iven context, at ./src/rcl/init.c:241
[ERROR] [listener-1]: process has died [pid 6204, exit code 1, cmd '/home/ayrton/Documents/bluerov/bl
uerov_ws/install/autonomous_rov/lib/autonomous_rov/listener --ros-args -r __node:=listener -r __ns:=/
bluerov2'].
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 launch autonomous_rov run_listener.lau
nch
```

BlueROV2 Camera



```
INFO
[listener-1]: This node is named listener
[listener-1]: namespace =/bluerov2
[listener-1]: Publishers created.
[listener-1]: ask router to create endpoint to enable
[listener-1]: disarming requested, wait_for_service
[listener-1]: Disarming Succeeded
[listener-1]: stream rate requested, wait_for_service
[listener-1]: set stream rate Succeeded
[listener-1]: Subscriptions done.
```

ayrton@ayrton-LOQ-15IAX9E: ~/Documents/bluerov/bluerov_ws 10x27

```
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102, in shutdown
rcl_shutdown already called on the given context,
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 run autonomous_rov video --ros-args -r
install/autonomous_rov/lib/autonomous_rov/video", line 130, in shutdown
shutdown(context=context)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/utilities.py", line 58, in shutdown
return context.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102, in shutdown
self.__context.shutdown()
rclpy._rclpy_pybind11.RCLLError: failed to shutdown: rcl_shutdown already called on the given context,
at ./src/rcl/init.c:241
[ros2run]: Process exited with failure 1
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 run autonomous_rov video --ros-args -r
__node:=video_nodes
```

```
rclpy.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/_init_.py", line 130, in shutdown
shutdown(context=context)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/utilities.py", line 58, in shutdown
return context.shutdown()
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/context.py", line 102, in shutdown
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rclpy._rclpy_pybind11.RCLLError: failed to shutdown: rcl_shutdown already called on the given context,
at ./src/rcl/init.c:241
[ros2run]: Process exited with failure 1
ayrton@ayrton-LOQ-15IAX9E:~/Documents/bluerov/bluerov_ws$ ros2 run autonomous_rov video --ros-args -r
__node:=video_nodes
```

